

HOW TO TEST WITH AGILITY

ALAN RICHARDSON

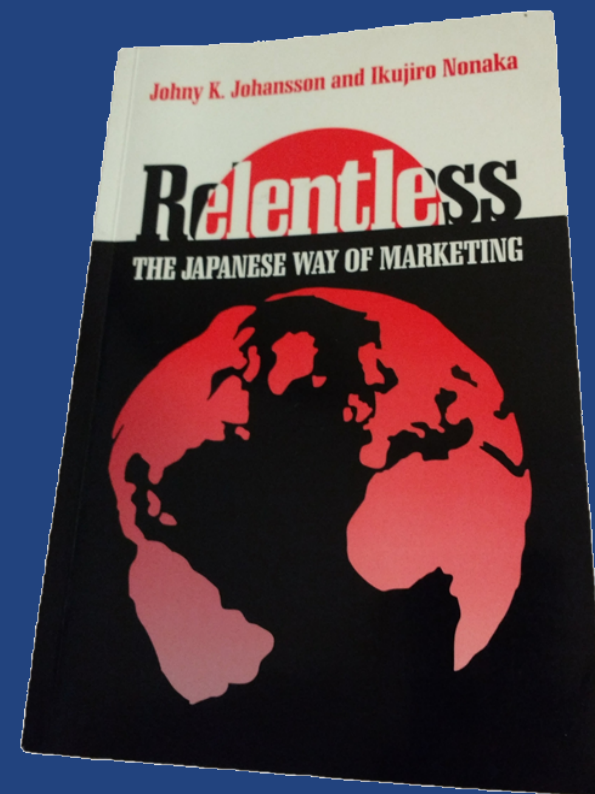
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CONFIDENCE
DO YOU KNOW HOW TO
'AGILE'?

CONFIDENCE COMES WITH EXPERIENCE

- > DO YOU KNOW HOW TO WORK WITH 'AGILITY'?
- > DO YOU KNOW MULTIPLE WAYS TO PERFORM A TASK?
 - > CAN YOU MAKE DECISIONS?
- > CAN YOU TAKE RESPONSIBILITY FOR THOSE DECISIONS?

RELENTLESS



RELENTLESS – THE JAPANESE WAY OF MARKETING BY JOHNNY K.
JOHANSSON AND IKUJIRO NONAKA, 1996

RELENTLESS

**'MARKETING BY THE JAPANESE. ... IS
FUNDAMENTALLY AN APPLICATION OF COMMON
SENSE. ...THEY PREFER**

- > SYNTHESIZING OVER ANALYZING.**
- > COMMON WORDS OVER PROFESSIONAL JARGON.**
- > AND SIMPLICITY OVER SOPHISTICATION.'**

AGILITY
TRUMPS AGILE
COMPLIANCE

'...MARKETING IS TOO IMPORTANT TO LEAVE TO THE PROFESSIONALS. CUSTOMERS WILL FALL THROUGH THE CRACKS BETWEEN SPECIALIZATIONS.'

RELENTLESS – THE JAPANESE WAY OF MARKETING BY JOHNY K. JOHANSSON AND IKUJIRO NONAKA, 1996, PG 7

TO UNDERSTAND HOW TO
'TEST' WITH AGILITY
WE FIRST NEED TO
UNDERSTAND 'TESTING'

WHAT DO WE
GET FROM
'TESTING'

NOT 'WHY DO WE TEST?'

WHAT DO WE GET FROM 'TESTING'

- > WHAT DO WE ACHIEVE BY TESTING?
- > WHAT DO WE GAIN FROM THE PROCESS OF TESTING?
 - > WHAT HAPPENS IF WE DO NOT TEST?
 - > ETC.

WHAT ARE THE OUTCOMES OF HAVING TESTED.

Q: WHAT DO WE GET FROM 'TESTING'

A: VARIES DEPENDING ON WHO YOU ASK

**I – CONFIDENCE I – BUGS I
I – KNOWLEDGE I – ASSURANCE I
I – DATA I – INFORMATION I
I – AUTOMATION I – COMPLIANCE I
I – RISK IDENTIFICATION I – RISK MITIGATION I
I – CHANGE IMPACT ASSESSMENT I – SLA CHECKING I
I – ETC. I I**

IF TESTING WAS QUESTIONING

- > HOW DO WE KNOW?
- > WHAT EVIDENCE DO WE HAVE?
 - > WHAT IF?
 - > WHAT ELSE?
 - > ETC.

TO ANSWER THOSE QUESTIONS

- > WE MODEL SYSTEMS
- > IDENTIFY AND EXPLORE RISK
- > USE APPROPRIATE COMMUNICATION
- > COMPARE MODELS TO SYSTEMS VIA EXPERIMENTATION AND EXPLORATION

... A BASIC TEST APPROACH

TEST AGILITY

- > CONTEXTUAL
- > FLEXIBILITY OF APPROACH
- > RESPONSIVENESS
- > ADD VALUE & AVOIDING WASTE
- > DOING/SAYING THE 'RIGHT' THING
- > ETC.

**TESTING WITH AGILITY
DOES NOT REQUIRE AN
AGILE PROJECT**

AGILE PROJECTS FORCE SOME AGILITY

- > REDUCED DOCUMENTATION
- > SMALLER TIMESCALES
- > LESS UP FRONT INFORMATION
- > TEST AND DEVELOP IN PARALLEL
 - > CO-LOCATED
 - > ETC.

AGILE RITUALS AND DEFINITIONS

- > 3 AMIGOS, STANDUPS, RETROSPECTIVE, STORY POINTS, STORIES, DEFINITION OF DONE
- > "AGILE", "DEVOPS", "AUTOMATION PYRAMID", "ATDD", "BDD", "TDD", "CONTINUOUS INTEGRATION", "CONTINUOUS DELIVERY", "SDET"
 - > ETC.

NOT REALLY A LOT ABOUT TESTING IN THERE

SOME SEEM TO BE ABOUT TESTING

BUT THEY ARE NOT REALLY.

I – DEFINITION OF DONE I – TEST DRIVEN DEVELOPMENT (TDD) I

I – ATDD I – BDD I

I – SDET I – CONTINUOUS INTEGRATION I

BUT THEY DO PROVIDE SOME OF THE GAINS WE ASSOCIATE WITH TESTING.

- E.G. CONFIDENCE, ASSURANCE, BUGS, CHANGE IMPACT ASSESSMENT, SLA CHECKING, AUTOMATION

TESTING IS NOT THE ONLY
WAY TO ACHIEVE THOSE
GAINS

SO HOW DOES TESTING FIT IN TO AGILE?

WORKING ON AGILE PROJECTS IS HARD WHEN WE TRY TO 'BE' AGILE

MAKE IT EASIER BY

- > WORKING FLEXIBLY
- > USE COMMONSENSE COMMUNICATION
 - > TRUST YOURSELF

RIGIDITY EXAMPLES

- > PARALYSIS
- > BEGINNERS KNOW THE RITUALS BUT EXHIBIT NO VARIETY
 - > FOLLOW THE RITUALS AND IT WILL ALL WORK OUT
 - > WAIT FOR THE COACH

**I LEARNED TO WORK WITH
AGILITY RATHER THAN
AGILE RITUALS AND
DEFINITIONS.**

**STICKING TO RITUALS AND
DEFINITIONS LED TO
RIGIDITY. RATHER THAN
AGILITY.**

**FOCUS ON
OUTCOMES
RATHER THAN PROCESSES**

WHAT IS AGILE SOFTWARE DEVELOPMENT... REALLY?

**REGULARLY RELEASE
SOFTWARE THAT WORKS
AND WHICH ADDS VALUE
WITH EACH RELEASE.**

SO HOW DO WE
TEST WITH
AGILITY?

HOW DO WE KNOW THAT WE... REGULARLY RELEASE SOFTWARE THAT WORKS AND WHICH ADDS VALUE WITH EACH RELEASE

- > HOW DO WE KNOW IT WORKS?
 - > WHAT DOES 'WORKS' MEAN?
- > HOW DO WE KNOW IT DOES WHAT IT SHOULD?
- > HOW DO WE KNOW IT DOESN'T DO WHAT IT SHOULDN'T?
 - > HOW DO WE KNOW IT ADDS VALUE?

**WE CAN TEST. TO FIND
ANSWERS TO THOSE
QUESTIONS.**

AUGMENT AGILE PROCESSES WITH TESTING

WHAT DOES AGILE SOFTWARE DEVELOPMENT LOOK LIKE?

- > GOALS – EPICS, RELEASE PLANNING
 - > STORIES
- > ACCEPTANCE CRITERIA
 - > DEVELOPMENT

HOW MIGHT TESTING FIT INTO THAT?

GOALS: EPICS, RELEASE PLANNING

> QUESTIONS, RISKS

STORIES

> QUESTIONS, RISKS,

> NOTES ON HOW TO TEST

HOW MIGHT TESTING FIT INTO THAT?

ACCEPTANCE CRITERIA

- > QUESTIONS, RISKS.
- > NOTES ON HOW TO TEST.
- > WHAT WE WILL COVER.
- > WHAT WE WILL NOT COVER

HOW MIGHT TESTING FIT INTO THAT?

DEVELOPMENT

- > ACCEPTANCE CRITERIA VALIDATION.
- > AUTOMATE ACCEPTANCE CRITERIA VALIDATION FOR CI.
- > EXPLORATORY TESTING (BASED ON RISK, QUESTIONS, WHAT WE LEARN FROM ACC CRITERIA VALIDATION, ETC.)

WHO DOES ALL OF THAT TESTING?

ANYONE. CAN...

- > QUESTIONS, RISKS, NOTES.
- > ACCEPTANCE CRITERIA VALIDATION

PEOPLE WITH TESTING SKILLS WILL DO IT
'DIFFERENTLY'

WHO DOES ALL OF THAT TESTING?

PEOPLE WHO CAN CODE, CAN...

- > AUTOMATE ACCEPTANCE CRITERIA VALIDATION FOR CI

PEOPLE WITH TESTING SKILLS CAN AUTOMATE TO
SUPPORT 'TESTING' NOT JUST 'VALIDATION'

WHO DOES ALL OF THAT TESTING?

PEOPLE WHO CAN TEST, CAN...

- > EXPLORATORY TESTING
- > TECHNICAL ACCEPTANCE CRITERIA VALIDATION (PERFORMANCE, SECURITY, ACCESSIBILITY)

AGILITY REQUIRES UTILISING SKILLS.

NOT CONFORMING TO ROLES.

WE NEED PEOPLE WHO CAN TEST WELL.

SOMEONE WHO CAN TEST
WELL. WILL SEE
OPPORTUNITIES TO TEST
THAT OTHER PEOPLE DO
NOT SEE.

**SOMEONE WHO CAN TEST
WELL. WILL TEST BETTER
THAN OTHER PEOPLE.**

**GIVE PEOPLE
THE TIME
TO APPLY THEIR SKILLS**

LET'S GET HUNG UP ON 'AUTOMATED TESTING'

THESE ARE PROCESS QUESTIONS

- > SHOULD WE AUTOMATE?
 - > HOW MUCH?
- > WHERE? API? GUI? – AVOID GUI ITS FLAKY!
 - > WHEN?

OUTCOME QUESTIONS

WHAT QUESTIONS DO WE WANT OUR 'AUTOMATED TESTING' TO ANSWER?

- > DOES THIS SOFTWARE CONTINUE TO WORK?
- > DOES THIS SOFTWARE CONTINUE TO MEET THE AGREED ACCEPTANCE CRITERIA?
- > ETC.

IF WE CARE ABOUT THE
ANSWER THEN WE DO
WHAT IT TAKES TO GET
THE ANSWER.

ROBUST AUTOMATED EXECUTION

- > MULTI-SKILLED
 - > COMMIT
 - > RE-USE THE AUTOMATED EXECUTION FOR EXPLORATION AS WELL AS VALIDATION
 - > PART OF YOUR PROJECT COST AND TIME
- SOMETIMES IT ISN'T WORTH THE EFFORT.

ACCEPTANCE CRITERIA VALIDATION IS NOT ENOUGH

- > HOW AMBIGUOUS IS THE ACCEPTANCE CRITERIA?
 - > DOES IT COVER RISK?
- > DOES IT COVER IMPACT ON PREVIOUS AREAS OF THE SYSTEM?
 - > ARE THE EXAMPLES 'COMPLETE'?

HOW DEEP DO YOU HAVE TO OBSERVE ACCEPTANCE CRITERIA?

- > HTTP TRAFFIC?
- > MEMORY CONSUMPTION?
- > PROCESSOR SPEED?
- > FILE USAGE?

EXPLORATORY AND TECHNICAL TESTING

- > WHAT IF?
- > WHAT ELSE?

USING THE RESULTS OF QUESTIONING AND
OBSERVATION TO IMPROVE OUR MODEL AND EXPAND
OUR SCOPE OF TESTING.

END TO END TESTING

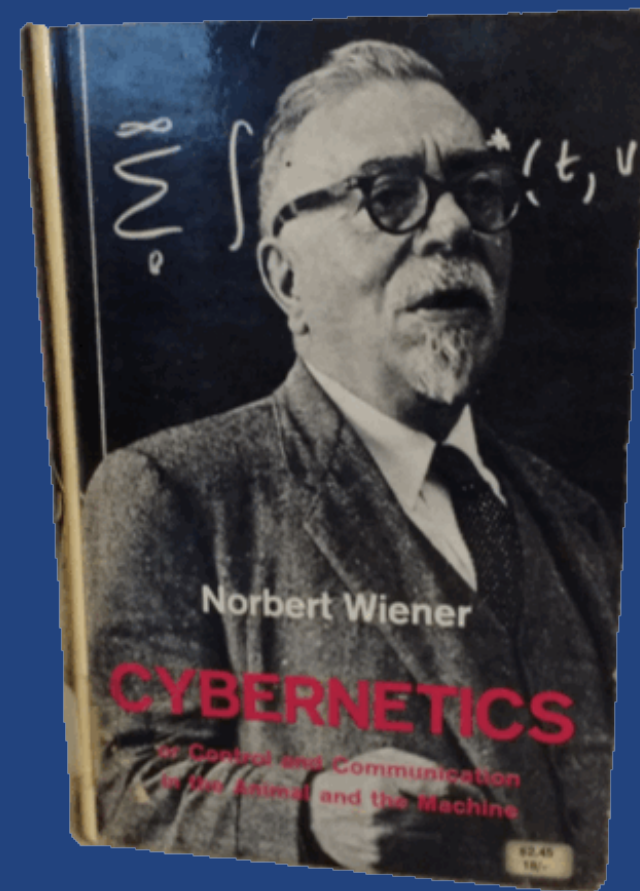
- > DO WE STILL NEED TO TEST END TO END WHEN WE WORK WITH AGILE?
- > UNIT TESTING IS DESIGN AND CONFORMANCE
- > AUTOMATED EXECUTION OF FEATURE FLOWS

WE NEED AGILITY

**WE CREATE A
CUSTOMISED AND
TAILORED PROCESS FOR
OUR ORGANISATION**

- > WE BUILD A SYSTEM **OF** DEVELOPMENT
- > WE WANT THE SYSTEM TO WORK EFFECTIVELY IN OUR ENVIRONMENT
 - > PEOPLE ARE PART OF THAT SYSTEM
- > WE AUTOMATE AS PART OF THAT SYSTEM
 - > WE TEST AS PART OF THAT SYSTEM

AGILITY – LESSONS LEARNED FROM CYBERNETICS

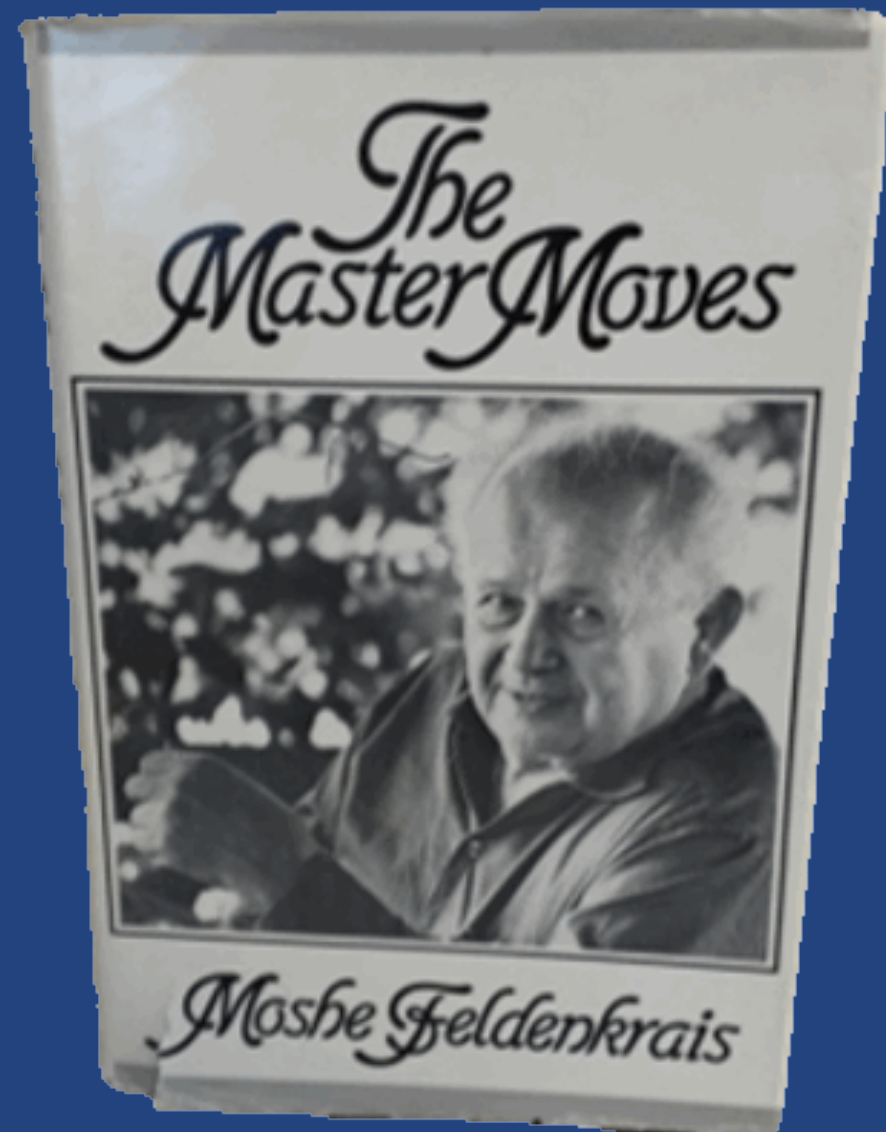


CYBERNETICS BY NORBERT WIENER – 1948 – [BIT.LY/2BA2JQV](https://bit.ly/2BA2JQV)

AGILITY – LESSONS LEARNED FROM CYBERNETICS

- > SCIENCE OF FEEDBACK AND CONTROL
 - > REQUISITE VARIETY
- > CYBERNETICS BY NORBERT WIENER – 1948 – [BIT.LY/2BA2JQV](https://bit.ly/2BA2JQV)
 - > SEE ALSO STAFFORD BEER, W. ROSS ASHBY

AGILITY – LESSONS LEARNED FROM MOSHE FELDENKRAIS



AGILITY – LESSONS LEARNED FROM MOSHE FELDENKRAIS

“...LEARNING MEANS HAVING AT LEAST ANOTHER WAY
OF DOING THE SAME THING.”

“TRY SOMETHING YOU KNOW AND LEARN TO DO IT
ANOTHER WAY.”

MOSHE FELDENKRAIS – THE MASTER MOVES, 1984

AGILITY – LESSONS LEARNED FROM MOSHE FELDENKRAIS

‘NOW, TAKE A MINUTE TO TRY FOR YOURSELVES
ANYTHING YOU WANT, TO SEE WHAT YOU COULD
IMPROVE IN YOUR MOVEMENT.’

MOSHE FELDENKRAIS – THE MASTER MOVES, 1984

INDIVIDUALLY: CAN YOU TRUST YOURSELF?

- > DO YOU UNDERSTAND THE TECHNOLOGIES BEING USED?
 - > DO YOU KNOW HOW TO TEST?
- > DO YOU KNOW HOW TO TEST THESE TECHNOLOGIES?
 - > WILL YOU SPEAK OUT?
 - > WILL YOU ASK QUESTIONS?

**DO YOU HAVE THE
NECESSARY SKILLS AND
ATTITUDE?**

CULTURALLY: CAN YOU TRUST YOURSELVES?

AGILITY REQUIRES TRUST IN YOURSELF AND YOUR TEAM.

- > ARE YOU TRUSTWORTHY?
 - > ARE YOU SKILLED?
- > ARE YOU EXPERIENCED?
- > ARE YOU FLEXIBLE?

WORKING ON AGILE PROJECTS IS HARD WHEN WE TRY TO 'BE' AGILE

MAKE IT EASIER BY

- > UNDERSTANDING OUTCOMES
 - > WORKING FLEXIBLY
- > USE COMMONSENSE COMMUNICATION
 - > TRUST YOURSELF

WORK WITH AGILITY

END

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BIO

ALAN IS A TEST CONSULTANT WHO WORKS AT A TECHNICAL LEVEL USING TECHNIQUES FROM PSYCHOTHERAPY AND COMPUTER SCIENCE. IN HIS SPARE TIME ALAN IS CURRENTLY PROGRAMMING A MULTI-USER TEXT ADVENTURE GAME, A TWITTER CLIENT AND SOME BUGGY JAVASCRIPT GAMES IN THE STYLE OF THE CASCADE CASSETTE 50. ALAN IS THE AUTHOR OF THE BOOKS 'DEAR EVIL TESTER', 'JAVA FOR TESTERS' AND 'AUTOMATING AND TESTING A REST API'. ALAN'S MAIN WEBSITE IS COMPENDIUMDEV.CO.UK AND